

Table 1: Joint strategy states and paired mean-Recall differences at $K = 20$. Percentages and percentage points.

Cell	Method	Mean R	VI	Always	Never	ΔR C0 [95% CI]
WANDS/MiniLM	Raw (seven schedules)	29.47	19.58	20.95	59.47	-1.34 [-2.53, -0.22]
WANDS/MiniLM	Canonical ascending	28.72	0.00	28.72	71.28	-2.09 [-4.03, -0.22]
WANDS/MiniLM	Canonical descending	29.56	0.00	29.56	70.44	-1.25 [-2.69, +0.06]
WANDS/MiniLM	Canonical type/style	29.22	0.00	29.22	70.78	-1.59 [-3.49, +0.22]
WANDS/MiniLM	Set-Mean	31.59	0.00	31.59	68.41	+0.78 [-1.24, +2.80]
WANDS/MiniLM	Centroid, $M = 2$	29.57	0.00	29.57	70.43	-1.24 [-2.84, +0.31]
WANDS/MiniLM	Multi-vector max, $M = 2$	29.77	0.00	29.77	70.23	-1.04 [-2.69, +0.55]
WANDS/MiniLM	BM25	34.72	0.00	34.72	65.28	+3.91 [+1.37, +6.44]
WANDS/MiniLM	Hybrid (seven schedules)	36.02	13.70	29.49	56.81	+5.21 [+3.45, +7.12]
WANDS/BGE	Raw (seven schedules)	36.04	11.93	30.47	57.60	-0.07 [-0.43, +0.28]
WANDS/BGE	Canonical ascending	36.07	0.00	36.07	63.93	-0.03 [-0.53, +0.45]
WANDS/BGE	Canonical descending	36.28	0.00	36.28	63.72	+0.17 [-0.72, +1.16]
WANDS/BGE	Canonical type/style	36.48	0.00	36.48	63.52	+0.38 [-0.08, +0.89]
WANDS/BGE	Set-Mean	35.30	0.00	35.30	64.70	-0.80 [-1.99, +0.32]
WANDS/BGE	Centroid, $M = 2$	36.40	0.00	36.40	63.60	+0.30 [-0.06, +0.66]
WANDS/BGE	Multi-vector max, $M = 2$	36.39	0.00	36.39	63.61	+0.29 [-0.06, +0.64]
WANDS/BGE	BM25	34.72	0.00	34.72	65.28	-1.38 [-3.77, +0.95]
WANDS/BGE	Hybrid (seven schedules)	38.07	7.52	34.30	58.18	+1.97 [+0.76, +3.28]
ESCI/MiniLM	Raw (seven schedules)	67.58	4.04	65.47	30.49	-0.38 [-0.83, +0.08]
ESCI/MiniLM	Canonical ascending	67.50	0.00	67.50	32.50	-0.46 [-0.98, +0.05]
ESCI/MiniLM	Canonical descending	67.52	0.00	67.52	32.48	-0.43 [-1.13, +0.30]
ESCI/MiniLM	Canonical type/style	67.48	0.00	67.48	32.52	-0.47 [-1.00, +0.03]
ESCI/MiniLM	Set-Mean	66.43	0.00	66.43	33.57	-1.53 [-3.08, +0.08]
ESCI/MiniLM	Centroid, $M = 2$	67.90	0.00	67.90	32.10	-0.06 [-0.59, +0.47]
ESCI/MiniLM	Multi-vector max, $M = 2$	67.69	0.00	67.69	32.31	-0.27 [-0.74, +0.22]
ESCI/MiniLM	BM25	67.69	0.00	67.69	32.31	-0.27 [-2.90, +2.38]
ESCI/MiniLM	Hybrid (seven schedules)	71.49	3.27	69.85	26.89	+3.54 [+1.66, +5.44]
ESCI/BGE	Raw (seven schedules)	73.83	3.66	71.89	24.44	-0.27 [-0.55, -0.02]
ESCI/BGE	Canonical ascending	73.84	0.00	73.84	26.16	-0.26 [-0.65, +0.09]
ESCI/BGE	Canonical descending	73.60	0.00	73.60	26.40	-0.51 [-0.95, -0.12]
ESCI/BGE	Canonical type/style	73.82	0.00	73.82	26.18	-0.29 [-0.68, +0.08]
ESCI/BGE	Set-Mean	71.98	0.00	71.98	28.02	-2.13 [-3.74, -0.57]
ESCI/BGE	Centroid, $M = 2$	73.86	0.00	73.86	26.14	-0.24 [-0.64, +0.10]
ESCI/BGE	Multi-vector max, $M = 2$	73.93	0.00	73.93	26.07	-0.17 [-0.48, +0.13]
ESCI/BGE	BM25	67.69	0.00	67.69	32.31	-6.42 [-8.93, -3.97]
ESCI/BGE	Hybrid (seven schedules)	74.84	2.31	73.69	24.00	+0.73 [-1.12, +2.56]

Table 2: Joint strategy states and paired mean-Recall differences at $K = 100$. Percentages and percentage points.

Cell	Method	Mean R	VI	Always	Never	ΔR C0 [95% CI]
WANDS/MiniLM	Raw (seven schedules)	56.61	21.89	45.15	32.96	-1.27 [-2.06, -0.52]
WANDS/MiniLM	Canonical ascending	56.05	0.00	56.05	43.95	-1.83 [-3.17, -0.64]
WANDS/MiniLM	Canonical descending	56.16	0.00	56.16	43.84	-1.72 [-2.97, -0.58]
WANDS/MiniLM	Canonical type/style	57.19	0.00	57.19	42.81	-0.70 [-2.12, +0.68]
WANDS/MiniLM	Set-Mean	58.81	0.00	58.81	41.19	+0.92 [-0.65, +2.52]
WANDS/MiniLM	Centroid, $M = 2$	56.44	0.00	56.44	43.56	-1.45 [-2.99, -0.11]
WANDS/MiniLM	Multi-vector max, $M = 2$	57.04	0.00	57.04	42.96	-0.84 [-2.14, +0.33]
WANDS/MiniLM	BM25	63.23	0.00	63.23	36.77	+5.35 [+2.72, +8.01]
WANDS/MiniLM	Hybrid (seven schedules)	65.55	11.08	60.16	28.76	+7.66 [+5.95, +9.51]
WANDS/BGE	Raw (seven schedules)	63.62	12.30	57.18	30.51	+0.45 [-0.33, +1.39]
WANDS/BGE	Canonical ascending	63.24	0.00	63.24	36.76	+0.07 [-1.47, +1.65]
WANDS/BGE	Canonical descending	63.61	0.00	63.61	36.39	+0.44 [-0.39, +1.47]
WANDS/BGE	Canonical type/style	63.56	0.00	63.56	36.44	+0.39 [-0.97, +1.86]
WANDS/BGE	Set-Mean	61.82	0.00	61.82	38.18	-1.35 [-3.25, +0.54]
WANDS/BGE	Centroid, $M = 2$	63.96	0.00	63.96	36.04	+0.78 [-0.09, +1.83]
WANDS/BGE	Multi-vector max, $M = 2$	64.00	0.00	64.00	36.00	+0.83 [+0.03, +1.84]
WANDS/BGE	BM25	63.23	0.00	63.23	36.77	+0.06 [-2.55, +2.65]
WANDS/BGE	Hybrid (seven schedules)	67.77	7.05	64.26	28.69	+4.60 [+3.01, +6.30]
ESCI/MiniLM	Raw (seven schedules)	86.19	1.41	85.53	13.06	+0.06 [-0.24, +0.43]
ESCI/MiniLM	Canonical ascending	86.41	0.00	86.41	13.59	+0.28 [-0.05, +0.76]
ESCI/MiniLM	Canonical descending	86.38	0.00	86.38	13.62	+0.25 [-0.22, +0.81]
ESCI/MiniLM	Canonical type/style	86.36	0.00	86.36	13.64	+0.22 [-0.09, +0.70]
ESCI/MiniLM	Set-Mean	84.50	0.00	84.50	15.50	-1.63 [-2.91, -0.37]
ESCI/MiniLM	Centroid, $M = 2$	86.33	0.00	86.33	13.67	+0.19 [-0.24, +0.72]
ESCI/MiniLM	Multi-vector max, $M = 2$	86.35	0.00	86.35	13.65	+0.22 [-0.21, +0.76]
ESCI/MiniLM	BM25	82.83	0.00	82.83	17.17	-3.30 [-5.48, -1.17]
ESCI/MiniLM	Hybrid (seven schedules)	89.57	0.56	89.31	10.13	+3.43 [+2.18, +4.82]
ESCI/BGE	Raw (seven schedules)	91.04	0.72	90.73	8.55	+0.06 [-0.05, +0.17]
ESCI/BGE	Canonical ascending	91.05	0.00	91.05	8.95	+0.07 [-0.04, +0.18]
ESCI/BGE	Canonical descending	91.01	0.00	91.01	8.99	+0.04 [-0.13, +0.19]
ESCI/BGE	Canonical type/style	91.09	0.00	91.09	8.91	+0.11 [-0.01, +0.23]
ESCI/BGE	Set-Mean	88.95	0.00	88.95	11.05	-2.03 [-3.17, -0.94]
ESCI/BGE	Centroid, $M = 2$	91.07	0.00	91.07	8.93	+0.09 [-0.08, +0.25]
ESCI/BGE	Multi-vector max, $M = 2$	91.09	0.00	91.09	8.91	+0.11 [-0.07, +0.28]
ESCI/BGE	BM25	82.83	0.00	82.83	17.17	-8.15 [-10.16, -6.25]
ESCI/BGE	Hybrid (seven schedules)	91.61	0.74	91.25	8.00	+0.63 [-0.16, +1.45]

Table 3: BGE fixed view-budget sweep. Each method has structural VI=0.

Dataset	Method	M	R@20	R@100	$\Delta R@100$ vs C0 [95% CI]
WANDS	centroid	2	36.40	63.96	+0.78 [-0.09, +1.83]
WANDS	max	2	36.39	64.00	+0.83 [+0.03, +1.84]
WANDS	centroid	4	36.44	64.39	+1.22 [+0.34, +2.30]
WANDS	max	4	36.51	63.86	+0.69 [-0.29, +1.84]
WANDS	centroid	7	36.35	64.16	+0.99 [+0.02, +2.11]
WANDS	max	7	36.46	64.15	+0.98 [-0.11, +2.22]
ESCI	centroid	2	73.86	91.07	+0.09 [-0.08, +0.25]
ESCI	max	2	73.93	91.09	+0.11 [-0.07, +0.28]
ESCI	centroid	4	73.86	91.01	+0.04 [-0.12, +0.19]
ESCI	max	4	73.98	91.05	+0.07 [-0.11, +0.25]
ESCI	centroid	7	73.70	91.01	+0.04 [-0.12, +0.18]
ESCI	max	7	73.97	91.05	+0.07 [-0.11, +0.25]

M=2 remains the common setting. M=4 and M=7 do not replace it. All intervals are recomputed using seed 2026091701. Centroid stores one vector per product; max stores M, retaining repeated views.

Table 4: Exact within-query cancellation, all six C0 contrasts at both cutoffs.

Dataset / encoder	Order	K	Equal > 0	All q	Changed q	All (%)	Changed (%)
WANDS / MiniLM	C1	20	57	308	211	18.51	27.01
WANDS / MiniLM	C2s1	20	52	308	207	16.88	25.12
WANDS / MiniLM	C2s2	20	60	308	198	19.48	30.30
WANDS / MiniLM	C2s3	20	59	308	195	19.16	30.26
WANDS / MiniLM	C2s4	20	64	308	196	20.78	32.65
WANDS / MiniLM	C2s5	20	67	308	202	21.75	33.17
WANDS / MiniLM	C1	100	21	308	199	6.82	10.55
WANDS / MiniLM	C2s1	100	32	308	193	10.39	16.58
WANDS / MiniLM	C2s2	100	30	308	196	9.74	15.31
WANDS / MiniLM	C2s3	100	22	308	193	7.14	11.40
WANDS / MiniLM	C2s4	100	39	308	193	12.66	20.21
WANDS / MiniLM	C2s5	100	22	308	192	7.14	11.46
WANDS / BGE	C1	20	73	308	187	23.70	39.04
WANDS / BGE	C2s1	20	78	308	179	25.32	43.58
WANDS / BGE	C2s2	20	78	308	184	25.32	42.39
WANDS / BGE	C2s3	20	68	308	181	22.08	37.57
WANDS / BGE	C2s4	20	69	308	182	22.40	37.91
WANDS / BGE	C2s5	20	67	308	182	21.75	36.81
WANDS / BGE	C1	100	25	308	168	8.12	14.88
WANDS / BGE	C2s1	100	30	308	159	9.74	18.87
WANDS / BGE	C2s2	100	31	308	163	10.06	19.02
WANDS / BGE	C2s3	100	40	308	155	12.99	25.81
WANDS / BGE	C2s4	100	32	308	148	10.39	21.62
WANDS / BGE	C2s5	100	42	308	162	13.64	25.93
WANDS / GTE	C1	20	61	308	191	19.81	31.94
WANDS / GTE	C2s1	20	67	308	195	21.75	34.36
WANDS / GTE	C2s2	20	81	308	198	26.30	40.91
WANDS / GTE	C2s3	20	76	308	193	24.68	39.38
WANDS / GTE	C2s4	20	70	308	193	22.73	36.27
WANDS / GTE	C2s5	20	76	308	189	24.68	40.21
WANDS / GTE	C1	100	37	308	163	12.01	22.70
WANDS / GTE	C2s1	100	31	308	171	10.06	18.13
WANDS / GTE	C2s2	100	29	308	170	9.42	17.06
WANDS / GTE	C2s3	100	32	308	171	10.39	18.71
WANDS / GTE	C2s4	100	33	308	165	10.71	20.00
WANDS / GTE	C2s5	100	43	308	169	13.96	25.44
ESCI / MiniLM	C1	20	14	499	92	2.81	15.22
ESCI / MiniLM	C2s1	20	14	499	73	2.81	19.18
ESCI / MiniLM	C2s2	20	10	499	68	2.00	14.71
ESCI / MiniLM	C2s3	20	8	499	60	1.60	13.33
ESCI / MiniLM	C2s4	20	7	499	80	1.40	8.75
ESCI / MiniLM	C2s5	20	13	499	74	2.61	17.57
ESCI / MiniLM	C1	100	2	499	30	0.40	6.67
ESCI / MiniLM	C2s1	100	2	499	20	0.40	10.00
ESCI / MiniLM	C2s2	100	2	499	16	0.40	12.50
ESCI / MiniLM	C2s3	100	2	499	19	0.40	10.53
ESCI / MiniLM	C2s4	100	1	499	19	0.20	5.26
ESCI / MiniLM	C2s5	100	2	499	17	0.40	11.76
ESCI / BGE	C1	20	16	499	78	3.21	20.51
ESCI / BGE	C2s1	20	12	499	55	2.40	21.82
ESCI / BGE	C2s2	20	14	499	67	2.81	20.90
ESCI / BGE	C2s3	20	16	499	63	3.21	25.40

Dataset / encoder	Order	K	Equal > 0	All q	Changed q	All (%)	Changed (%)
ESCI / BGE	C2s4	20	12	499	66	2.40	18.18
ESCI / BGE	C2s5	20	12	499	62	2.40	19.35
ESCI / BGE	C1	100	1	499	25	0.20	4.00
ESCI / BGE	C2s1	100	3	499	17	0.60	17.65
ESCI / BGE	C2s2	100	0	499	17	0.00	0.00
ESCI / BGE	C2s3	100	1	499	20	0.20	5.00
ESCI / BGE	C2s4	100	0	499	20	0.00	0.00
ESCI / BGE	C2s5	100	2	499	21	0.40	9.52
ESCI / GTE	C1	20	21	499	123	4.21	17.07
ESCI / GTE	C2s1	20	19	499	96	3.81	19.79
ESCI / GTE	C2s2	20	17	499	102	3.41	16.67
ESCI / GTE	C2s3	20	19	499	92	3.81	20.65
ESCI / GTE	C2s4	20	19	499	104	3.81	18.27
ESCI / GTE	C2s5	20	15	499	104	3.01	14.42
ESCI / GTE	C1	100	4	499	49	0.80	8.16
ESCI / GTE	C2s1	100	3	499	35	0.60	8.57
ESCI / GTE	C2s2	100	3	499	38	0.60	7.89
ESCI / GTE	C2s3	100	3	499	35	0.60	8.57
ESCI / GTE	C2s4	100	3	499	44	0.60	6.82
ESCI / GTE	C2s5	100	1	499	39	0.20	2.56